# *Employee Attrition Analysis – Executive Summary*

This analysis examined data from *1,470 employees* to identify patterns associated with employee attrition and to help HR proactively identify employees who may be at risk of leaving. The overall attrition rate in the organization was *16.12%*, meaning approximately *1 in every 6 employees* left the company.

The analysis found that attrition is not evenly distributed across the organization. The *Sales department* recorded the highest attrition rate (*20.63%*), followed by *Human Resources (19.05%)*, while *Research & Development* had the lowest attrition *(13.84%)*. Among individual job roles, *Sales Representatives* experienced the highest attrition *(39.76%)*, followed by *Laboratory Technicians (23.94%)* and *Human Resources employees (23.08%)*. Employees within their *first two years at the company* were also significantly more likely to leave than longer-serving employees.

A predictive machine learning model was developed to identify employees who may be at risk of leaving. Among the three models evaluated, *Logistic Regression* provided the best balance between prediction quality and interpretability. It successfully identified approximately *66% of employees who were likely to leave*, making it the most practical model for supporting HR retention efforts. Although another model achieved higher overall accuracy, it identified far fewer employees who were actually at risk, making it less suitable for this business problem.

The model also revealed that *working overtime*, *frequent business travel*, and *certain job roles were stronger indicators of employee attrition than salary alone*. While employees who left generally earned lower salaries on average, compensation was not among the model's strongest predictors. This suggests that workload, travel requirements, and the nature of specific roles have a greater influence on employee retention than pay alone.

Based on these findings, HR should prioritize retention efforts for employees who *frequently work overtime, travel regularly*, or belong to *high-risk roles such as Sales Representatives and Laboratory Technicians, particularly during their first two years of employment*. Regular check-ins, mentoring programs, workload management, career development opportunities, and initiatives that improve work-life balance may help reduce voluntary attrition.

Finally, this model should be *viewed as a decision-support tool rather than a decision-making tool*. It identifies employees who may be at higher risk of leaving based on historical data, but it cannot determine the personal or organizational reasons behind an individual's decision to resign. HR should therefore use these predictions alongside manager feedback, employee engagement surveys, and professional judgment when planning retention strategies